

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: Mo-99			FORMS: Soluble		
PHYSICAL CHARACTERISTICS					
HALF-LIFE: 66 hours					
DECAY EMISSIONS					
Includes Tc-99m emissions under the assumption of secular equilibrium					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV, Ave)	%	E (keV)	%
141	89	443	82		
740	12	133	16		
181	6	119*	9		
18	6	20*	4		
-Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY					
C – level (low hazard): ≤ 20 mCi					
B – level (moderate hazard): > 20 mCi, ≤ 1 Ci					
A – level (high hazard): > 1 Ci					
EXTERNAL RADIATION HAZARDS			INTERNAL RADIATION HAZARDS		
Includes Tc-99m emissions			Annual Limit on Intake: 1000 µCi (Ingestion)		
Gamma dose rate, point source at 1 ft, 1 mCi:			1000 µCi (Inhalation)		
1.54 mrem/h					
Beta dose rate to skin, point source at 1 ft, 1 mCi:			The values above indicate the activity taken into the		
400 mrem/h			body that would result in either 5 rem to the whole		
Contamination skin dose, uniform deposit of 1 µCi per cm²:			body (CEDE) or 50 rem to an organ or tissue (CDE).		
7000 mrem/h					
SHIELDING			DOSIMETRY AND BIOASSAY REQS		
Includes Tc-99m emissions			Whole-body and finger-ring dosimeters are required		
Gammas/X-rays:			for handling 5 mCi or more, or 1 mCi amounts		
2 cm of lead will reduce the gamma dose rate by 90%.			weekly. Urine assays may be required after large		
Betas/electrons:			spills or contaminations.		
4 mm of plastic will absorb all emissions.					
Bremsstrahlung may be created and require					
additional shielding.					
SPECIAL PROBLEMS AND PRECAUTIONS:					
1. Recommended survey probe: PGM or NaI					
2. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often.					
3. Survey work areas before, during, and after work. Work areas may require shielding to keep dose ALARA. Instrument and smear surveys are required.					
4. Segregate waste to those with half-lives of between 1 and 8 days. Survey the waste disposal area to ensure exposure rates are less than 2 mR/hr at 1 foot.					
5. Limit soluble waste to the sewer to less than 100 µCi/day per lab.					

References:

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- Johnson, T.E., Birkby, B.K. (2012). Health Physics and Radiological Health (4th ed.). Baltimore, MD: Lippincott Williams & Wilkins.
- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
- Peplow, D. (2020) Specific Gamma-Ray Dose Constants with Current Emission Data. *Health Physics*, 118(4):402-416; 2020.
- Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
- 10.CFR.20 – Standards for Protection Against Radiation (2019). Retrieved from <https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/>